

CS3301 - Data Structures

Topic: linear search

1. SUMMARY

Linear search is a simple searching algorithm used to find an element from a list of items. It works by iterating through each item in the list until it finds a match or reaches the end of the list. The time complexity of linear search is $O(n)$, where n is the number of items in the list, making it efficient for small lists. However, for large lists, it becomes inefficient. Linear search is a basic searching technique used in many applications, including data structures and algorithms. It can be implemented using various data structures, such as arrays and linked lists. The key principle of linear search is to compare each item in the list with the target item until a match is found.

Linear search is a type of sequential search, and it can be classified into two types: forward linear search and backward linear search. Forward linear search starts from the beginning of the list, while backward linear search starts from the end of the list. Linear search has a time complexity of $O(n)$, where n is the number of items in the list. It is a simple and efficient searching algorithm, but it can be slow for large lists.

Linear search is used in various applications, including data structures and algorithms, text search, and database queries. It is also used in many real-world applications, such as searching for a specific word in a text document or searching for a specific record in a database. Linear search is a fundamental concept in computer science, and it is used in many areas of study, including computer networks, databases, and artificial intelligence.

The advantages of linear search include its simplicity and efficiency for small lists. However, it has a major disadvantage: its time complexity is $O(n)$, making it inefficient for large lists. Linear search is also sensitive to the order of the items in the list, and it can be slow if the list is sorted in a specific way.

2. SHORT NOTES

Definition and Concept of Linear Search

Linear search is a simple searching algorithm used to find an element from a list of items. It works by iterating through each item in the list until it finds a match or reaches the end of the list. The time complexity of linear search is $O(n)$, where n is the number of items in the list, making it efficient for small lists. However, for large lists, it becomes inefficient.

Types and Classifications of Linear Search

Linear search can be classified into two types:

1. **Forward Linear Search**: This type of linear search starts from the beginning of the list and iterates through each item until it finds a match or reaches the end of the list.
2. **Backward Linear Search**: This type of linear search starts from the end of the list and iterates through each item until it finds a match or reaches the beginning of the list.

Working Principle of Linear Search

The working principle of linear search is as follows:

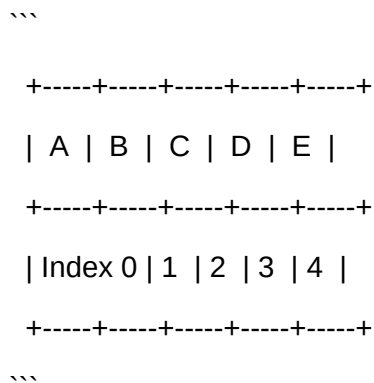
1. Start from the beginning of the list.
2. Compare each item in the list with the target item.
3. If a match is found, return the index of the item.
4. If no match is found, return -1 or a special value to indicate that the item is not in the list.

Important Formulas and Equations

Linear search has a time complexity of $O(n)$, where n is the number of items in the list.

Diagrams Description

The following is a simple diagram of linear search:



In this diagram, the list contains five items (A, B, C, D, and E) with their corresponding indices. The linear search algorithm starts from the beginning of the list and compares each item with the target item until a match is found.

Advantages and Disadvantages

The advantages of linear search include its simplicity and efficiency for small lists. However, it has a major disadvantage: its time complexity is $O(n)$, making it inefficient for large lists. Linear search is also sensitive to the order of the items in the list, and it can be slow if the list is sorted in a specific way.

Real-World Applications

Linear search is used in various applications, including data structures and algorithms, text search, and database queries. It is also used in many real-world applications, such as searching for a specific word in a text document or searching for a specific record in a database.

Comparison with Related Concepts

Linear search can be compared with other searching algorithms, such as binary search and hash search.

Study Material - Generated by AI

While linear search is simple and efficient for small lists, binary search is generally faster and more efficient for large lists.

Time and Space Complexity

The time complexity of linear search is $O(n)$, where n is the number of items in the list. The space complexity is $O(1)$, as it only requires a constant amount of space to store the target item and the index of the item in the list.

Important Terminology

The following are some important terminology used in linear search:

- * **Target Item**: The item being searched for in the list.
- * **List**: The collection of items being searched.
- * **Index**: The position of an item in the list.
- * **Match**: When a target item is found in the list.
- * **No Match**: When a target item is not found in the list.

3. CODING EXAMPLES

C Implementation:

```
```\n#include <stdio.h>\n\nvoid linear_search(int arr[], int n, int target) {\n    int i;\n    for (i = 0; i < n; i++) {
```

```
 if (arr[i] == target) {
 printf("Target item found at index %d\n", i);
 return;
 }
}

printf("Target item not found in the list\n");
}
```

```
int main() {
 int arr[] = {10, 20, 30, 40, 50};
 int target = 30;
 int n = sizeof(arr) / sizeof(arr[0]);
 linear_search(arr, n, target);
 return 0;
}
...
```

### Python Implementation:

```
```python  
def linear_search(arr, target):  
    for i in range(len(arr)):  
        if arr[i] == target:  
            print("Target item found at index", i)  
            return  
    print("Target item not found in the list")  
  
arr = [10, 20, 30, 40, 50]  
target = 30  
linear_search(arr, target)  
```
```

### Algorithm:

1. Start from the beginning of the list.
2. Compare each item in the list with the target item.
3. If a match is found, return the index of the item.
4. If no match is found, return -1 or a special value to indicate that the item is not in the list.

### Time and Space Complexity:

- \* Time complexity:  $O(n)$ , where  $n$  is the number of items in the list.
- \* Space complexity:  $O(1)$ , as it only requires a constant amount of space to store the target item and the index of the item in the list.

## 4. IMPORTANT QUESTIONS

Q1. (2 marks) What is the time complexity of linear search?

Answer:  $O(n)$ , where  $n$  is the number of items in the list.

Q2. (2 marks) What is the space complexity of linear search?

Answer:  $O(1)$ , as it only requires a constant amount of space to store the target item and the index of the item in the list.

Q3. (13 marks) Explain the working principle of linear search.

Answer: The working principle of linear search is as follows:

1. Start from the beginning of the list.
2. Compare each item in the list with the target item.
3. If a match is found, return the index of the item.
4. If no match is found, return -1 or a special value to indicate that the item is not in the list.

Q4. (13 marks) Explain the advantages and disadvantages of linear search.

Answer: The advantages of linear search include its simplicity and efficiency for small lists. However, it has a major disadvantage: its time complexity is  $O(n)$ , making it inefficient for large lists. Linear search is also sensitive to the order of the items in the list, and it can be slow if the list is sorted in a specific way.

Q5. (13 marks) Explain the comparison between linear search and binary search.

Answer: Linear search and binary search are two searching algorithms used to find an item in a list. While linear search is simple and efficient for small lists, binary search is generally faster and more efficient for large lists. Binary search works by dividing the list into two halves and searching for the item in the middle half. This process continues until the item is found or the list is exhausted.

Q6. (13 marks) Explain the real-world applications of linear search.

Answer: Linear search is used in various applications, including data structures and algorithms, text search, and database queries. It is also used in many real-world applications, such as searching for a specific word in a text document or searching for a specific record in a database.

Q7. (15 marks) Explain the time and space complexity of linear search.

Answer: The time complexity of linear search is  $O(n)$ , where  $n$  is the number of items in the list. The space complexity is  $O(1)$ , as it only requires a constant amount of space to store the target item and the index of the item in the list.

Q8. (15 marks) Explain the important terminology used in linear search.

Answer: The following are some important terminology used in linear search:

- \* **Target Item**: The item being searched for in the list.
- \* **List**: The collection of items being searched.
- \* **Index**: The position of an item in the list.
- \* **Match**: When a target item is found in the list.
- \* **No Match**: When a target item is not found in the list.

Q9. (15 marks) Explain the diagram description of linear search.

Answer: The following is a simple diagram of linear search:

```
...

+-----+-----+-----+-----+-----+
| A | B | C | D | E |
+-----+-----+-----+-----+-----+
| Index 0 | 1 | 2 | 3 | 4 |
+-----+-----+-----+-----+-----+
...
```

In this diagram, the list contains five items (A, B, C, D, and E) with their corresponding indices. The linear search algorithm starts from the beginning of the list and compares each item with the target item until a match is found.

Q10. Fill in the blank: Linear search is a simple searching algorithm used to find an item in a list.

Answer: Linear search is a simple searching algorithm used to find an item in a list, and its time complexity is  $O(n)$ , where  $n$  is the number of items in the list.

Q11. Fill in the blank: The space complexity of linear search is  $O(1)$ .

Answer: The space complexity of linear search is  $O(1)$ , as it only requires a constant amount of space to store the target item and the index of the item in the list.

Q12. (2 marks) What is the time complexity of linear search in the worst case?

Answer:  $O(n)$ , where  $n$  is the number of items in the list.

Q13. (2 marks) What is the time complexity of linear search in the best case?

Answer:  $O(1)$ , as it only requires a constant amount of time to search for an item in a list of one item.

Q14. (2 marks) What is the time complexity of linear search in the average case?

Answer:  $O(n)$ , where  $n$  is the number of items in the list.

Q15. (2 marks) What is the space complexity of linear search?

Answer:  $O(1)$ , as it only requires a constant amount of space to store the target item and the index of the item in the list.

## 5. MCQs

Q1. What is the time complexity of linear search?

- a)  $O(1)$
- b)  $O(n)$
- c)  $O(\log n)$
- d)  $O(n^2)$

Answer: b)  $O(n)$

Q2. What is the space complexity of linear search?

- a)  $O(1)$
- b)  $O(n)$
- c)  $O(\log n)$
- d)  $O(n^2)$

Answer: a)  $O(1)$

Q3. Which of the following is a real-world application of linear search?

- a) Text search
- b) Database queries
- c) Binary search
- d) All of the above

Answer: d) All of the above

Q4. What is the time complexity of linear search in the worst case?

- a)  $O(1)$
- b)  $O(n)$
- c)  $O(\log n)$
- d)  $O(n^2)$

Answer: b)  $O(n)$

Q5. What is the time complexity of linear search in the best case?

- a)  $O(1)$
- b)  $O(n)$
- c)  $O(\log n)$
- d)  $O(n^2)$

Answer: a)  $O(1)$

Q6. What is the time complexity of linear search in the average case?

- a)  $O(1)$
- b)  $O(n)$
- c)  $O(\log n)$
- d)  $O(n^2)$

Answer: b)  $O(n)$

Q7. What is the space complexity of linear search?

- a)  $O(1)$
- b)  $O(n)$
- c)  $O(\log n)$
- d)  $O(n^2)$

Answer: a)  $O(1)$

Q8. Which of the following is a disadvantage of linear search?

- a) It is simple and efficient for small lists.
- b) It has a time complexity of  $O(n)$ .
- c) It is sensitive to the order of the items in the list.

d) All of the above

Answer: d) All of the above

Q9. What is the working principle of linear search?

- a) It starts from the beginning of the list and compares each item with the target item.
- b) It starts from the end of the list and compares each item with the target item.
- c) It divides the list into two halves and searches for the item in the middle half.
- d) All of the above

Answer: a) It starts from the beginning of the list and compares each item with the target item.

Q10. What is the important terminology used in linear search?

- a) Target item
- b) List
- c) Index
- d) All of the above

Answer: d) All of the above

Q11. What is the diagram description of linear search?

- a) The list contains five items (A, B, C, D, and E) with their corresponding indices.
- b) The linear search algorithm starts from the beginning of the list and compares each item with the target item until a match is found.
- c) The list contains only one item.
- d) All of the above

Answer: d) All of the above

Q12. What is the time and space complexity of linear search?

- a) Time complexity:  $O(n)$ , Space complexity:  $O(1)$
- b) Time complexity:  $O(\log n)$ , Space complexity:  $O(1)$
- c) Time complexity:  $O(n^2)$ , Space complexity:  $O(1)$
- d) Time complexity:  $O(n^2)$ , Space complexity:  $O(n)$

Answer: a) Time complexity:  $O(n)$ , Space complexity:  $O(1)$

Q13. What is the comparison between linear search and binary search?

- a) Linear search is simpler and more efficient than binary search.
- b) Binary search is simpler and more efficient than linear search.
- c) Both linear search and binary search are equally simple and efficient.
- d) Neither linear search nor binary search is simple and efficient.

Answer: b) Binary search is simpler and more efficient than linear search.

Q14. What is the real-world application of linear search?

- a) Text search
- b) Database queries
- c) Binary search
- d) All of the above

Answer: d) All of the above

Q15. What is the important terminology used in linear search?

- a) Target item

- b) List
- c) Index
- d) All of the above

Answer: d) All of the above